

Team 02 – ResuMind

AI

Resume Processing Pipeline: Converting unstructured PDF documents into structured JSON data using AI and intelligent text processing



Meet Our Team

ResuMind AI brings together six specialized modules, each designed to handle a critical stage in the resume processing pipeline. From initial file upload to final validation, every team member ensures data quality and accuracy.



Pratik

AWS Deployment & Data Transformation



Akanksha

Mathematical Matching



Divya

Advisor Workflow



Sayali

Analysis and Output



Vikrant

The Backend & Storage Layer



Umaina

Schema Validation

PDF
document



Problem Statement

The Challenge

Resume data arrives in PDF format—unstructured, inconsistent, and impossible for AI systems to process directly. Each document follows different layouts, contains unique formatting quirks, and presents information in unpredictable ways.

The Solution

Our pipeline transforms raw PDF documents into clean, structured JSON format that AI systems can understand and utilize. This conversion enables automated resume screening, skills matching, and candidate evaluation at scale.

1

Intelligent Data Extraction

2

Semantic Vector Matching

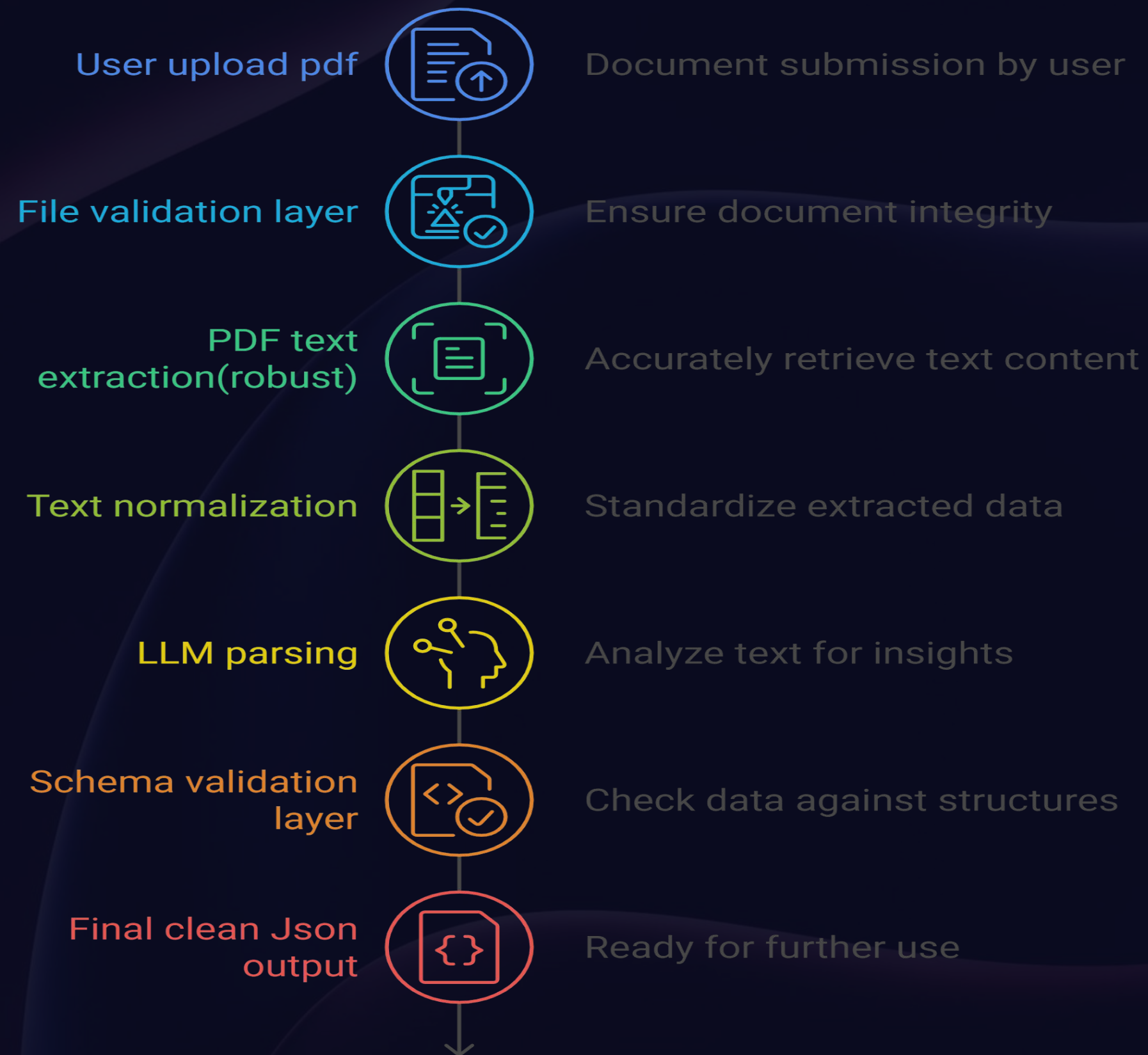
3

AI-Driven Gap Analysis

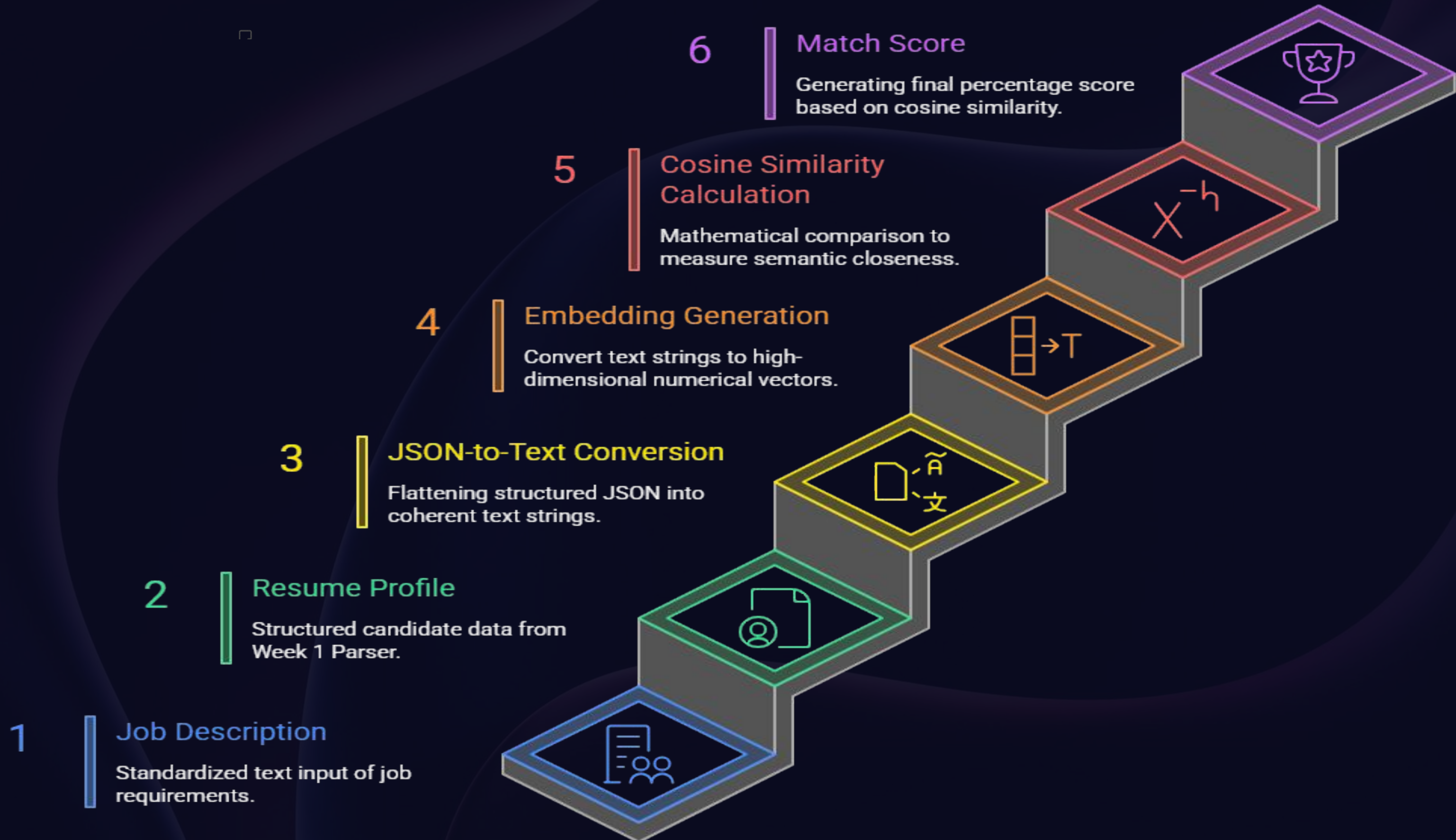
4

**Bulk Processing & HR Analytics
(The Portal)**

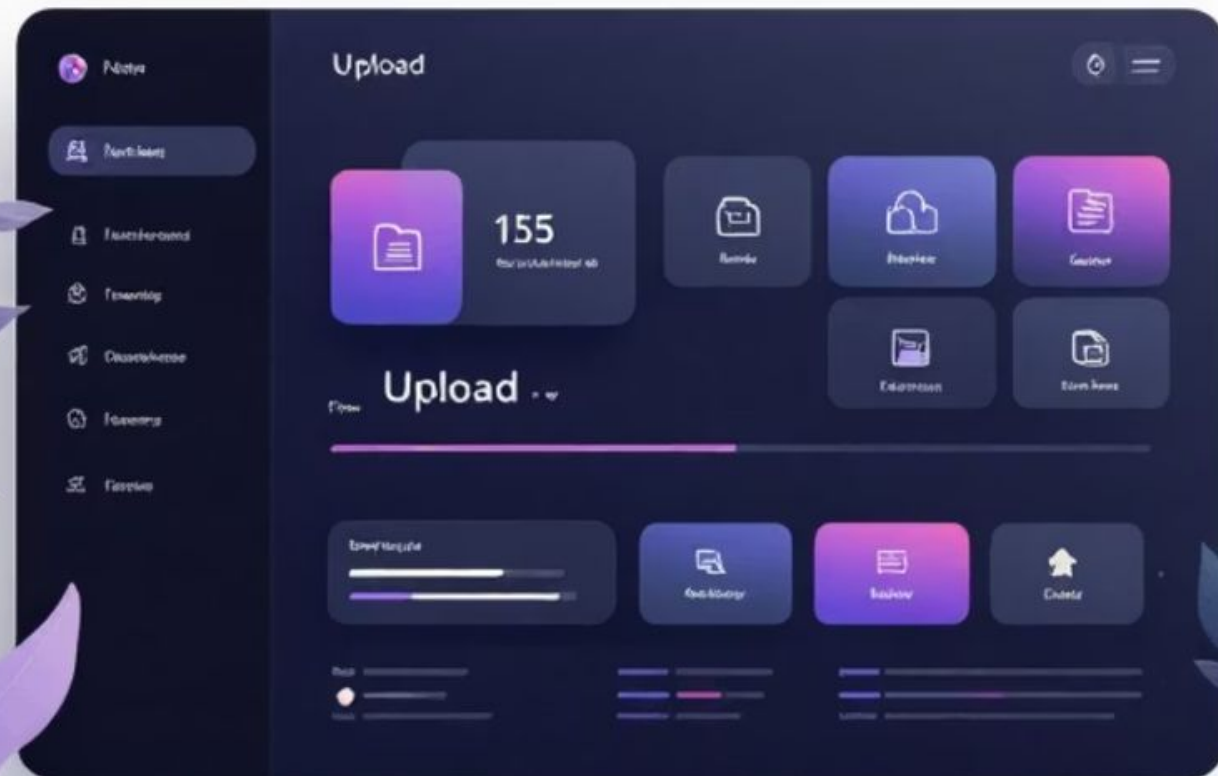
Week 1 : The Parser (Data Extraction)



Week 2 : The Matcher (Semantic Scoring)



Pratik & Akanksha – The Matcher (Semantic Scoring)



The Data Transformation

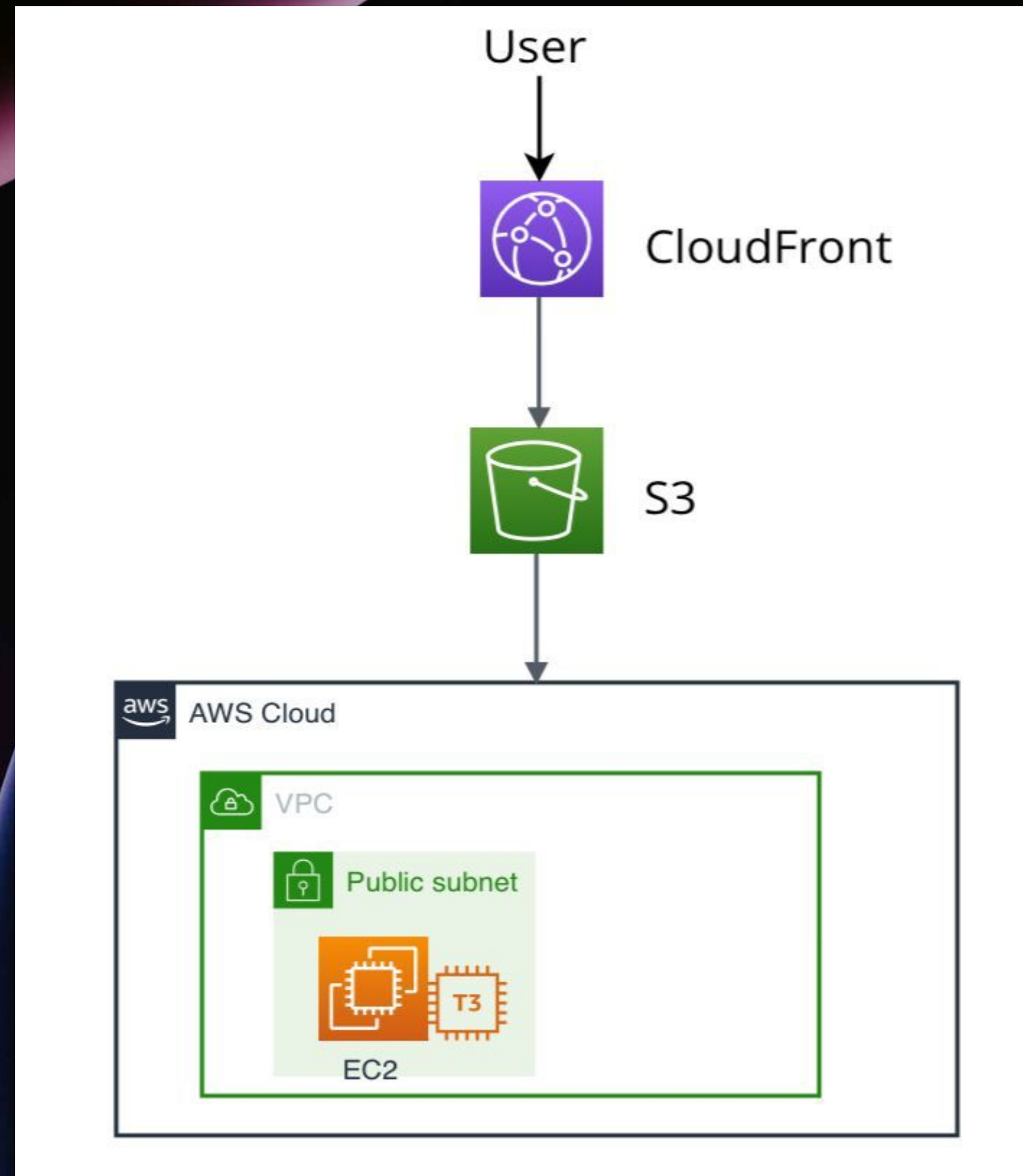
- **Text Normalization:**
Cleans and standardizes resume and job description text for consistency across formats
- **JSON-to-Text Conversion:**
Converts structured resume data into a simple, readable text format for AI processing
- **Embedding Generation:**
Transforms text into high-dimensional vectors to capture semantic meaning using LLMs

The Mathematical Matching

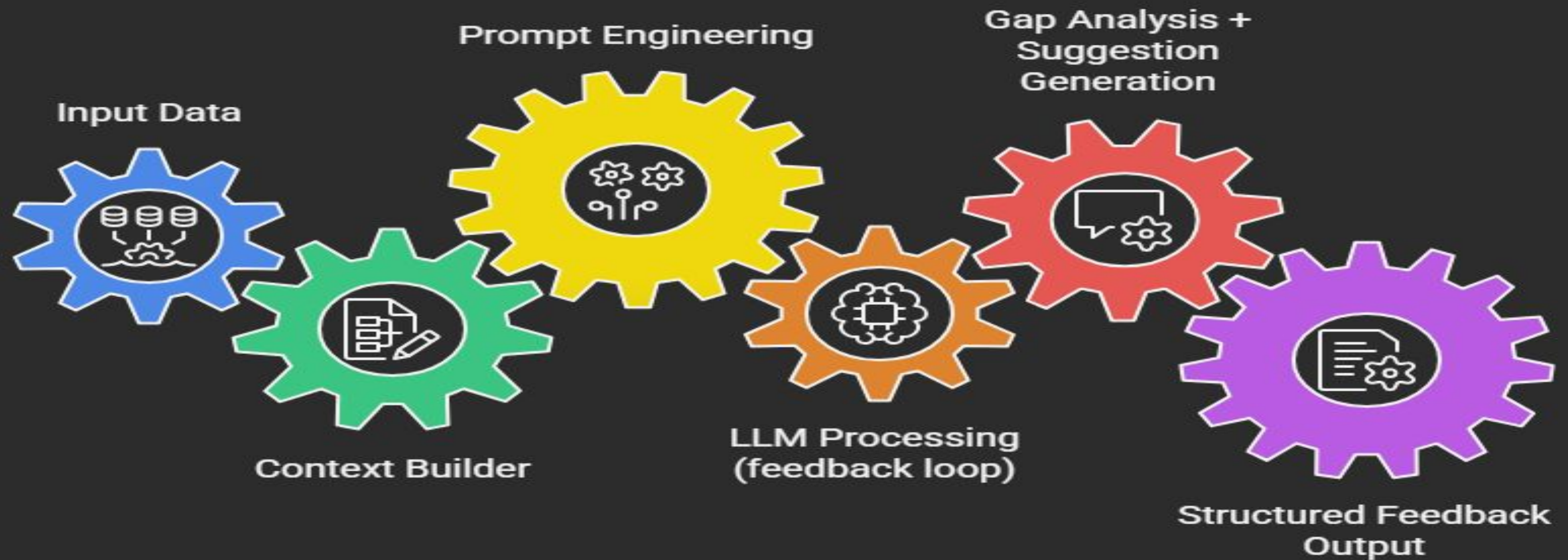
- **Vector Mapping:** Plots resume and job description as vectors in a multi-dimensional space
- **Cosine Similarity Logic:** Measures similarity by calculating the angle between vectors
- **Final Match Scoring:** Computes a percentage score (0%–100%) based on semantic similarity

AWS Deployment Architecture of ResuMind AI

Designed & Implemented by: Pratik Dashore



Week 3: The Advisor



Sayali & Divya – The Advisor

PRCCENT

1. That dilution has crept in and is less than the claim in the form

1. Two hundred thirty six boys have been tested so far and the results are as follows: 100 boys are normal, 100 are defective, and 36 are still to be tested.

2. The artificial alarm on the tower of selectivity will be the same as the one used on the other tower.

1. To include more than one register of action in boys in the case that conditions are the same in respect of the place in which the injury took place and the circumstances of the action.

2. The discipline that *Edinburgh* has inherited has been its own best defender, just as it has been the best ally of its cause elsewhere. They have made the discipline their characteristic and its intellectualism. It must thus be understood by the student that too many must have different educational experiences than he has to help all to reach the same level of their professional careers.

[illegible]

1. The yucca treatment (the first one) is a monthly liquid nitrogen and nitrogen dioxide gas treatment system. The second is a nitrogen and nitrogen dioxide gas treatment system. In the third and fourth treatments the first two are the same, but the third is a nitrogen and nitrogen dioxide gas treatment system.

2. The standards for general health assessment of fishes for export and for the treatment of fish to improve its value for export legislation are made by way of law like the results of the other laws but are not clear.

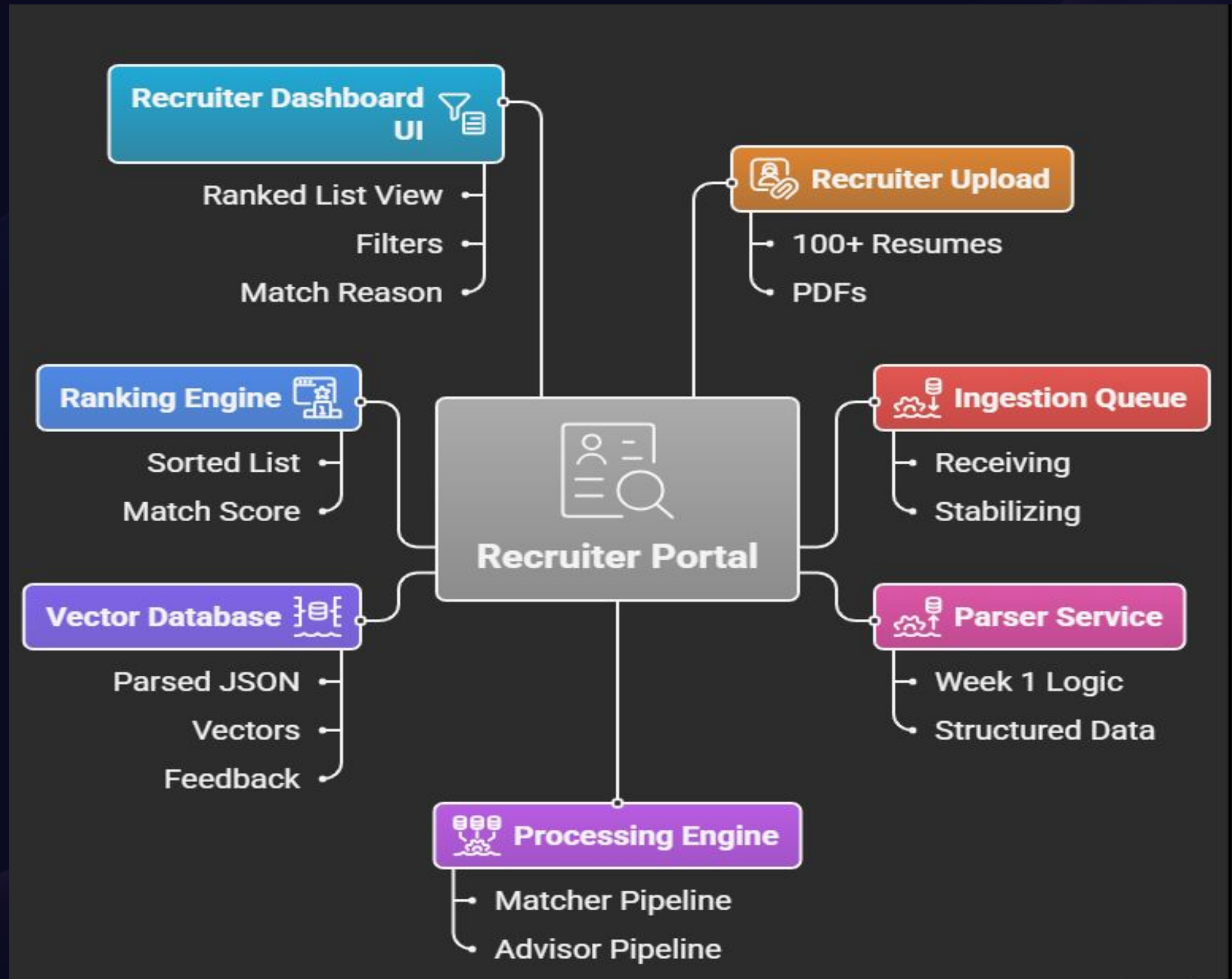
Advisor Workflow

- Takes input data from user/system
- Builds context by organizing important data
- Uses prompt engineering to guide AI
- Processes data using LLM (feedback loop)
- Improves results through iteration (repeat process)

Analysis & Output

- Performs gap analysis to find missing information
- Generates suggestions and improvements
- Provides structured feedback output
- Helps in better decision making
- Increases accuracy and efficiency

Week 4 : The Portal (Recruiter View)



Umaina & Vikrant – The Portal

The Backend & Storage Layer

- **Bulk File Ingestion:** Handles the high-speed upload of 100+ PDF resumes into the system.
- **File Storage Management:** Routes uploaded documents into a secure cloud storage layer for processing.
- **Parallel Processing Engine:** Triggers the Parser and Matcher simultaneously to handle multiple resumes at once.

The HR Dashboard & Ranking

- **Database Integration:** Stores the final structured JSON, match scores, and AI feedback in a vector database.
- **Ranking Engine Logic:** Sorts candidates automatically based on their calculated match percentage.
- **Recruiter UI Dashboard:** Displays the final table with filters and a clear "Match Reason" for every applicant.

Thank You